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Advances in Basic and Applied Sciences In Digital Age: Few Selected Areas

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CHAPTER 1

Applications of Graph Theory in Detection of Illegal Weapons

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ABSTRACT

Shipping is the most important means of universal transport. We all know that millions of containers are shipped from different docks all over the world. It has been estimated that around one million containers containing cargoes imported and exported through Kolkata. With this huge amount of goods being shipped, one of the major concerns is that some people may hide illegal items with cargo. In this project we apply graph theory and basic computational topology to identify a container with unusual characteristics. In this project, we consider different attributes like destination, company, goods and its weight along with the container. Our aim is to measure the changes in these attributes and use them to conclude whether a container need to be checked or not. The standard lengths of a containers are 40, 45 and 53 feet. The gross mass of one of these containers can range up to 40,000 kg. We will examine the dimensions and weight of this container. If there is any significant difference, we conclude that the container contents may be suspicious and hence such containers will get separated for further examinations.

Keywords: Graph, Directed Graph, Docks, Port SEZ (Special Economic Zone) Import, Export

INTRODUCTION

Graph theory is a branch of discrete mathematics. In mathematics and computer science, graph theory is the study of graphs which are mathematical structures used to model pair wise relations. Using graph many problems^[1,2] of real world can be represented. This paper is based on an application of graph theory for detecting the illegal weapons or drugs which can be shipped from different docks.

Given that 90% of India's foreign trade^[3] by volume takes place through water transportation. It has been estimated that around one million containers containing different cargoes imported and exported through Kolkata. West Bengal exported to more than 121 countries and imported from more than 68 countries^[5]. During 2019-2020 India handled total cargo around 1320 million tones. More than 3189 ships are handled only in Kolkata in 2020-2021.

With this huge amount of goods being shipped, one of the major concerns is that some people may hide illegal items with cargo. Hence It is imperative that India should have a world class port complying with all international standards.

According to India's Intelligence Bureau (IB) *Jihadi* terror groups smuggles terror apparatus to India because of anti-terror stand of India in the UN. The officials say that 70-75 percent of global cargo is shipped in containers which could be successfully used by terrorist organization to ship weapons of mass destruction into India.

So, to resolve these issues we need new technology which is use to enhance the transportation security^[4]. In this project we use graph theory and basic computational topology to identify a container with usual characteristics. There are some attributes we can use to get decision about the containers that it need to be check or not.

Objective of paper

The standard technology used for container monitoring is based on X-ray radiography. There are many drawbacks of current security system. It is very costly and a time taking process. It is difficult for X-ray machine to detect drugs and low-density materials. So, our goal is to use graph theory and basic computational topology to identify a container with unusual characteristics. There are some attributes we can use to get decision about the containers that it need to be check or not. In this project we consider different attributes like destination, company, goods and its weight along with the container.

Attributes

Containers: The two most important properties of a containers are its dimension and weight. The standard length of containers are 40ft, 45ft, 48ft, 53ft and mass of one container can range up to 40,000 kg. So, we will focus on dimension and weight of containers. If the weight of container is differed by more than a fix range then we take the weight of side joining port and good to be more.

Goods: There are many goods are shipped and some of the goods can be of similar weight as the weight of goods then we can think that there are some goods which container may contain illegal weapons and weight do not change by a fix amount. Like shipping Guns with iron ores has more chances than shipping with vegetable oil.

Destination: It also depends on destination like some countries where illegal activities are more then we need to check that containers.

Company: There are some companies which work with security then in that case we can ignore the checking of the containers. Now here we firstly model the problem and represent it using

graph where vertices are attributes and edge connected two vertices according to data given below in tables. Then we assign some weights to every edges. The weights depend on different factors like for a company its weight depend on its previous records. For importing and exporting countries the weights depend on the current situations, previous history etc. of those countries and for goods the weights depend on that which illegal weapons or drugs can be easily transported with that product.

Data About some ports and Docks

(I) *Shyama Prasad Mukherjee Port*



Import	Export	Import	Export	Destinations	2020 – 21
Fertilizer	Therma Coal	884	2359	China	5592
Coal	Iron Ore	7990	3179	Singapore	
Rice	Fly ash	5	3184	USA	
Limestone	Sand	3228	78	UAE	
Iron & Steel	Project Cargo	169	9	Japan	
Vegetable Oil	Coal Tar Sheet	2831	12	Malaysia	
Pulse/Peas		161		South Korea	
Sand		187			

(ii) Kolkata Dock System

Graph representing the destinations and goods import or export at SMP



Import	Export	Import (19-20)	Export (19-20)	Destination	Year	Number
Fertilizer	Iron ore	747	32	China	2016-17	1356
Rice	Steel Scarp	5		USA	2017-18	1356
Calcite Chips	Fly Ash	33	2020	Singapore	2018-19	1384
Petroleum Coke	Rice	99		Japan	2019-20	1292
Iron and Steel	Coal Tar Pitch	34	12	Germany	2020-21	1049
Machinery	Sand	8	78			
Timber	Iron and Steel	74		UK		
Car		1				
Cement		90	60	South Korea		

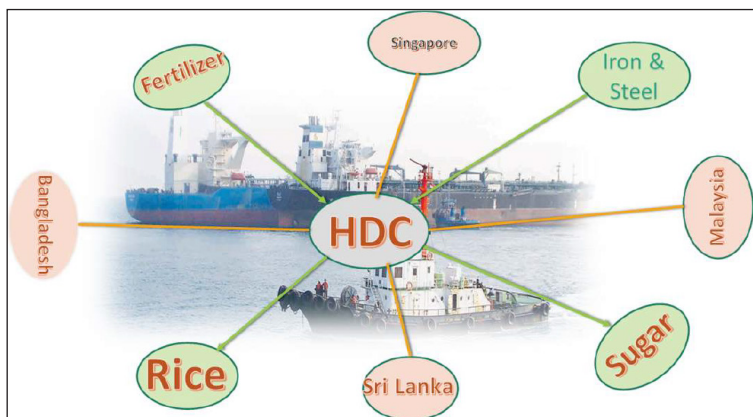
(iii) Haldia Docks Complex

Graph representing the destinations and goods import or export at KDC

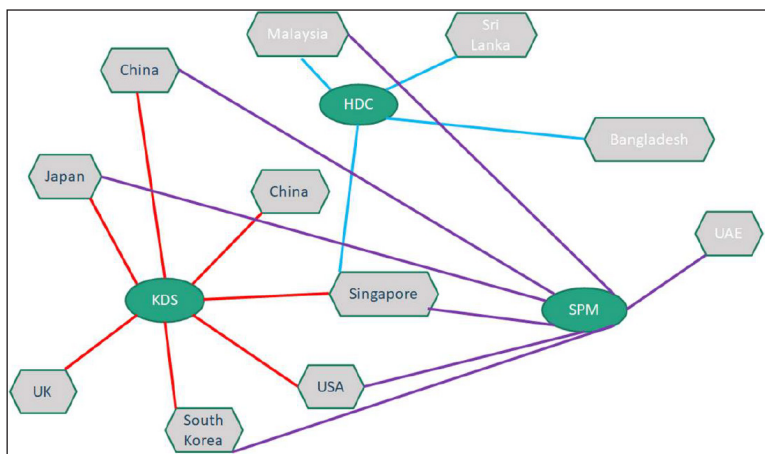


Goods		Container (2019-20)		Number of Ships		Destination
Import	Export	Import	Export	Year	Number	
Fertilizer	Thermal Coal	83623	85700	2016-17	2075	Bangladesh
Rock Phosphate	Iron Ore			2017-18	2315	Malaysia
Sulfur	Iron Steel			2018-19	2265	Sri Lanka
Salt	Pig Iron			2019-20	2221	Singapore
Coking Coal	Fly Ash			2020-21	2140	
Lime Stone	Rice					
Iron & Steel	Sugar					
Soda Ash	Veg. Oil					
Gypsum						
Dolomite						

Graph representing the destinations and goods import or export at HDC



Graph Representing connections between Different attributes:



Name of different companies working in Kolkata:

1. Res Trans- Logis, CIT Road, Kolkata
2. Roy Shipping Agency, Motilal Colony, Kolkata
3. Star Container Services, West Road, Kolkata

Size and Weight of different container

Container Length (ft)	Weight (kg)
20	21727
40	26780
45	29050
48	32650

Process:

Suppose W_{ij} be the weight of sides using i^{th} vertex and j^{th} vertices. Here for concluding that a container needs to check or not, we need to combine the weights of all the sides joining the port, destination and their corresponding container and goods. Here we use convex combination to combine different attributes. So, we use;

$$R = \lambda_1 w_{i1} + \lambda_2 w_{i2} + \lambda_3 w_{i3} + \lambda_4 w_{i4}$$

Where, $\sum_{i=1}^4 \lambda_i = 1$

If $R > 0.30$ then we conclude that the container needs to be check otherwise we can ignore the checking.

We use here convex combination because by using this we give the priority to different attributes, like if we give high priority to change in volume of container then their λ is greater than λ for other attributes. Here λ_i for different sides is depends on previous record and their corresponding priority.

- λ_1 stands for change in volume of container and $\lambda_1 = 0.4321$
- λ_2 stands for destination and $\lambda_2 = 0.3210$
- λ_3 stands for the goods which is shipping and $\lambda_3 = 0.1542$
- λ_4 stands for company and $\lambda_4 = 0.0927$

Like if we use Kolkata Dock System and rice is shipped to Singapore in a container of length 46ft and the weight difference is about 500kg and shipped by worldwide Cargo company then suppose their corresponding weight are 0.234, 0.124, 0.152, 0.672 then;

$$R = \lambda_1 w_{i1} + \lambda_2 w_{i2} + \lambda_3 w_{i3} + \lambda_4 w_{i4}$$

$$R = 0.4321*0.152 + 0.3210*0.124 + 0.1542*0.234 + 0.0927*0.672$$

$$R = 0.2038$$

Here $R < 0.30$ so this container needs not to be check.

FUTURE SCOPE

The accuracy of this model depends on weight of different sides joining different attributes.

- Here I used three ports of Kolkata and four attributes so we will increase the number of ports and attributes by this we will improve the accuracy of this model because as we increase the number of attributes the accuracy will increase.
- In this model we used random number for different values of λ but in real it depends on priority of different attributes so we will use information from port and by that we get better value of λ .
- We can also use the previous record of company to get weight of side joining different vertices to company name.
- We will use the real data for w_{ij} of different goods in different container so that we will measure the accurate difference in weight.

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